Contents

Chapter 1

Spin vs Kramers restricted MP2

1.1 Spin-unrestricted non-relativistic MP2 energy

The non-relativistic MP2 energy in spin-orbitals, assuming Einstein summation, is

$$E_{\text{MP2}} = \frac{1}{4}\bar{t}^{ij}_{ab}\bar{g}^{ab}_{ij}. \tag{1.1}$$

Here, $\bar{g}^{ab}_{ij}$ denotes the antisymmetrized electron repulsion integrals,

$$\bar{g}^{ab}_{ij} = g^{ab}_{ij} - g^{ab}_{ji} = \langle ij|ab\rangle - \langle ji|ab\rangle = \langle ij\|ab\rangle, \tag{1.2}$$

$$g^{ab}_{ij} = \langle ij|ab\rangle = \iint \phi^*_i(\mathbf{r1})\phi^*_j(\mathbf{r2})\frac{1}{|\mathbf{r1}-\mathbf{r2}|}\phi_a(\mathbf{r1})\phi_b(\mathbf{r2})d\mathbf{r1}d\mathbf{r2} \tag{1.3}$$

and

$$\bar{t}^{ij}_{ab} = \frac{\bar{g}^{ij}_{ab}}{\epsilon_i + \epsilon_j - \epsilon_a - \epsilon_b} = \frac{\bar{g}^{ij}_{ab}}{\Delta^{ij}_{ab}} \tag{1.4}$$

are the antisymmetrized MP2 amplitudes, where ϵ_p are the orbital energies. Indices i and j denote occupied orbitals, while indices a and b denote virtual orbitals. Note:

$$g^{ab}_{ij} = g^{ba}_{ji} = g^{*ij}_{\ \ ab} = g^{*ji}_{\ \ ba} \tag{1.5}$$

Before we look at spin-separated $E_{\mathbf{MP2}}$, we need to realize that not all spin-combinations are non-zero. This can be understood by working out the spin-block separation of $\bar{g}^{ab}_{ij}$:

$$\bar{g}^{ab}_{ij} = \bar{g}^{a_\alpha b_\alpha}_{i_\alpha j_\alpha} + \bar{g}^{a_\beta b_\beta}_{i_\beta j_\beta} + \bar{g}^{a_\alpha b_\beta}_{i_\alpha j_\beta} + \bar{g}^{a_\beta b_\alpha}_{i_\beta j_\alpha} + \bar{g}^{a_\alpha b_\beta}_{i_\beta j_\alpha} + \bar{g}^{a_\beta b_\alpha}_{i_\alpha j_\beta}$$

$$\bar{g}^{a_\alpha b_\alpha}_{i_\alpha j_\alpha} = g^{a_\alpha b_\alpha}_{i_\alpha j_\alpha} - g^{a_\alpha b_\alpha}_{j_\alpha i_\alpha} \tag{1.6}$$

$$\text{and} \quad \bar{g}^{a_\alpha b_\beta}_{i_\alpha j_\beta} = g^{a_\alpha b_\beta}_{i_\alpha j_\beta} - 0 \tag{1.7}$$

$$(g^{a_\alpha b_\beta}_{j_\beta i_\alpha} \text{ is zero due to violation of column spin equality}) \tag{1.8}$$

Decomposing the energy expression into its individual spin components,

$$E_{\text{MP2}} = E^{\alpha\alpha}_{\alpha\alpha} + E^{\beta\beta}_{\beta\beta} + E^{\alpha\beta}_{\alpha\beta} + E^{\beta\alpha}_{\beta\alpha} + E^{\alpha\beta}_{\beta\alpha} + E^{\beta\alpha}_{\alpha\beta} \tag{1.9}$$

$$= \frac{1}{4}\left[\bar{t}^{i_\alpha j_\alpha}_{a_\alpha b_\alpha}\bar{g}^{a_\alpha b_\alpha}_{i_\alpha j_\alpha} + \bar{t}^{i_\beta j_\beta}_{a_\beta b_\beta}\bar{g}^{a_\beta b_\beta}_{i_\beta j_\beta} + \bar{t}^{i_\alpha j_\beta}_{a_\alpha b_\beta}\bar{g}^{a_\alpha b_\beta}_{i_\alpha j_\beta} + \bar{t}^{i_\beta j_\alpha}_{a_\beta b_\alpha}\bar{g}^{a_\beta b_\alpha}_{i_\beta j_\alpha} + \bar{t}^{i_\alpha j_\beta}_{a_\beta b_\alpha}\bar{g}^{a_\beta b_\alpha}_{i_\alpha j_\beta} + \bar{t}^{i_\beta j_\alpha}_{a_\alpha b_\beta}\bar{g}^{a_\alpha b_\beta}_{i_\beta j_\alpha}\right] \tag{1.10}$$

Note that $E^{\alpha\beta}_{\alpha\beta} = E^{\beta\alpha}_{\beta\alpha}$ in Eq. (1.10). This can be seen by considering the fourth term in Eq. (1.10), which, upon swapping the columns of both $\bar{t}$ and $\bar{g}$ and then swapping the dummy indices, becomes identical to the third term in Eq. (1.10). Similarly, it can also be seen that $E^{\alpha\beta}_{\beta\alpha} = E^{\beta\alpha}_{\alpha\beta}$. Consequently, Eq. (1.10) can be rewritten as

$$E_{\mathrm{MP2}} = \frac{1}{4}\bar{t}^{i_\alpha j_\alpha}_{a_\alpha b_\alpha}\bar{g}^{a_\alpha b_\alpha}_{i_\alpha j_\alpha} + \frac{1}{4}\bar{t}^{i_\beta j_\beta}_{a_\beta b_\beta}\bar{g}^{a_\beta b_\beta}_{i_\beta j_\beta} + \frac{1}{2}\bar{t}^{i_\alpha j_\beta}_{a_\alpha b_\beta}\bar{g}^{a_\alpha b_\beta}_{i_\alpha j_\beta} + \frac{1}{2}\bar{t}^{i_\alpha j_\beta}_{a_\beta b_\alpha}\bar{g}^{a_\beta b_\alpha}_{i_\alpha j_\beta}. \tag{1.11}$$

Now consider the tensor network for $E^{\alpha\alpha}_{\alpha\alpha}$ only:

$$E^{\alpha\alpha}_{\alpha\alpha} = \frac{1}{4}\bar{t}^{i_\alpha j_\alpha}_{a_\alpha b_\alpha}\bar{g}^{a_\alpha b_\alpha}_{i_\alpha j_\alpha} \tag{1.12}$$

$$= \frac{1}{4}(t^{i_\alpha j_\alpha}_{a_\alpha b_\alpha} - t^{i_\alpha j_\alpha}_{b_\alpha a_\alpha})(g^{a_\alpha b_\alpha}_{i_\alpha j_\alpha} - g^{a_\alpha b_\alpha}_{j_\alpha i_\alpha}) \tag{1.13}$$

$$= \frac{1}{4}[t^{i_\alpha j_\alpha}_{a_\alpha b_\alpha} g^{a_\alpha b_\alpha}_{i_\alpha j_\alpha} - t^{i_\alpha j_\alpha}_{a_\alpha b_\alpha} g^{a_\alpha b_\alpha}_{j_\alpha i_\alpha} - t^{i_\alpha j_\alpha}_{b_\alpha a_\alpha} g^{a_\alpha b_\alpha}_{i_\alpha j_\alpha} + t^{i_\alpha j_\alpha}_{b_\alpha a_\alpha} g^{a_\alpha b_\alpha}_{j_\alpha i_\alpha}]. \tag{1.14}$$

The last term in Eq. (1.14) is equal to the first term, which can be seen by swapping the column indices of g and then exchanging the dummy indices a_α and b_α. An analogous relation holds between the second and third terms in Eq. (1.14). Therefore, the expression for $E_{\alpha\alpha}$ can be written in a compact form as

$$E^{\alpha\alpha}_{\alpha\alpha} = \frac{1}{2}[t^{i_\alpha j_\alpha}_{a_\alpha b_\alpha} g^{a_\alpha b_\alpha}_{i_\alpha j_\alpha} - t^{i_\alpha j_\alpha}_{a_\alpha b_\alpha} g^{a_\alpha b_\alpha}_{j_\alpha i_\alpha}] \tag{1.15}$$

$$= \frac{1}{2}t^{i_\alpha j_\alpha}_{a_\alpha b_\alpha}(g^{a_\alpha b_\alpha}_{i_\alpha j_\alpha} - g^{a_\alpha b_\alpha}_{j_\alpha i_\alpha}). \tag{1.16}$$

Similarly, the expression for $E^{\beta\beta}_{\beta\beta}$ can be written in compact form as

$$E^{\beta\beta}_{\beta\beta} = \frac{1}{2}t^{i_\beta j_\beta}_{a_\beta b_\beta}(g^{a_\beta b_\beta}_{i_\beta j_\beta} - g^{a_\beta b_\beta}_{j_\beta i_\beta}). \tag{1.17}$$

Now consider $2E^{\alpha\beta}_{\alpha\beta}$:

$$2E^{\alpha\beta}_{\alpha\beta} = \frac{1}{2}\bar{t}^{i_\alpha j_\beta}_{a_\alpha b_\beta}\bar{g}^{a_\alpha b_\beta}_{i_\alpha j_\beta} \tag{1.18}$$

$$= \frac{1}{2}(t^{i_\alpha j_\beta}_{a_\alpha b_\beta} - t^{i_\alpha j_\beta}_{b_\beta a_\alpha})(g^{a_\alpha b_\beta}_{i_\alpha j_\beta} - g^{a_\alpha b_\beta}_{j_\beta i_\alpha}). \tag{1.19}$$

The second term in each bracket vanishes because they violate column spin equivalence, and hence $2E^{\alpha\beta}_{\alpha\beta}$ reduces to

$$2E^{\alpha\beta}_{\alpha\beta} = \frac{1}{2}t^{i_\alpha j_\beta}_{a_\alpha b_\beta} g^{a_\alpha b_\beta}_{i_\alpha j_\beta}. \tag{1.20}$$

Similarly for $2E^{\alpha\beta}_{\beta\alpha}$, the first terms in the antisymmetrization brackets vanish and we are left with

$$2E^{\alpha\beta}_{\beta\alpha} = \frac{1}{2}t^{i_\alpha j_\beta}_{a_\alpha b_\beta} g^{a_\alpha b_\beta}_{i_\alpha j_\beta}. \tag{1.21}$$

Therefore, substituting the results of Eqs. (1.16), (1.17), (1.20) and (1.21) into Eq. (1.11), we get the expression for the total unrestricted MP2 energy as:

$$E_{\mathrm{UMP2}} = \frac{1}{2}t^{i_\alpha j_\alpha}_{a_\alpha b_\alpha}(g^{a_\alpha b_\alpha}_{i_\alpha j_\alpha} - g^{a_\alpha b_\alpha}_{j_\alpha i_\alpha}) + \frac{1}{2}t^{i_\beta j_\beta}_{a_\beta b_\beta}(g^{a_\beta b_\beta}_{i_\beta j_\beta} - g^{a_\beta b_\beta}_{j_\beta i_\beta}) + t^{i_\alpha j_\beta}_{a_\alpha b_\beta} g^{a_\alpha b_\beta}_{i_\alpha j_\beta} \tag{1.22}$$

1.2 Spin-restricted non-relativistic MP2 energy

The spin-restricted expression from MP2 energy can be obtained from Eq. (1.22) by simply dropping the spin indices as

$$\begin{aligned} E_{\mathrm{RMP2}} &= \frac{1}{2} t^{ij}_{ab} (g^{ab}_{ij} - g^{ab}_{ji}) + \frac{1}{2} t^{ij}_{ab} (g^{ab}_{ij} - g^{ab}_{ji}) + t^{ij}_{ab} g^{ab}_{ij} && (1.23) \\ &= t^{ij}_{ab} (2 g^{ab}_{ij} - g^{ab}_{ji}) && (1.24) \end{aligned}$$

1.3 Kramers unrestricted relativistic MP2 energy

If i denotes a spinor (Kramers up), then let $\bar{i}$ denote the time reversed version (Kramers down) of i, i.e.,

$$|i\rangle = \mathcal{T} |\bar{i}\rangle \tag{1.25}$$

where $\mathcal{T}$ is the time reversal operator.

$$\mathcal{T} = \begin{cases} i\boldsymbol{\sigma_y} \hat{K}_0, & 2 - component \\ i\boldsymbol{\Sigma_y} \hat{K}_0, & 4 - component \end{cases} \tag{1.26}$$

and $\hat{K}_0$ is the complex conjugation operator.

Due the symmetry of the matrix elements under time reversal, the following relationships arise for electron repulsion integrals for closed shell systems:

$$g^{rs}_{pq} = g^{*\bar{r}\bar{s}}_{\bar{p}\bar{q}} = g^{*\bar{r}q}_{\bar{p}s} = g^{*p\bar{s}}_{r\bar{q}} \tag{1.27}$$

$$-g^{*sp}_{\bar{q}r} = -g^{*ps}_{r\bar{q}} = g^{\bar{s}r}_{qp} = g^{r\bar{s}}_{pq} = -g^{*\bar{q}\bar{p}}_{s\bar{r}} = -g^{*\bar{p}\bar{q}}_{\bar{r}s} = g^{*q\bar{r}}_{\bar{s}\bar{p}} = g^{*\bar{r}q}_{\bar{p}\bar{s}} \tag{1.28}$$

$$g^{*pq}_{\bar{r}\bar{s}} = g^{\bar{r}\bar{s}}_{pq} \tag{1.29}$$

$$g^{*r\bar{s}}_{\bar{p}q} = g^{\bar{r}s}_{p\bar{q}} \tag{1.30}$$